

DTP

Debug and Test Ports

Beta release | Open-source synthesizable RTL

HIGHLIGHTS

- IEEE 1149.1-2013 JTAG TAP and boundary-scan infrastructure
- IEEE 1687-2014 iJTAG instrument networks with segment insertion bits
- Configurable secondary TAP topology for local and downstream chiplet access
- JTAG-to-AXI paths for SMC fabric, SMC OTP, and SEP OTP debug access
- Cross-trigger routing between chiplet-local and external trigger endpoints
- Lifecycle-controlled disable inputs for scan and memory-mapped debug paths
- Integrated debug clock-stop, boot-stall, and multi-slice reset controls

INTEGRATION FIT

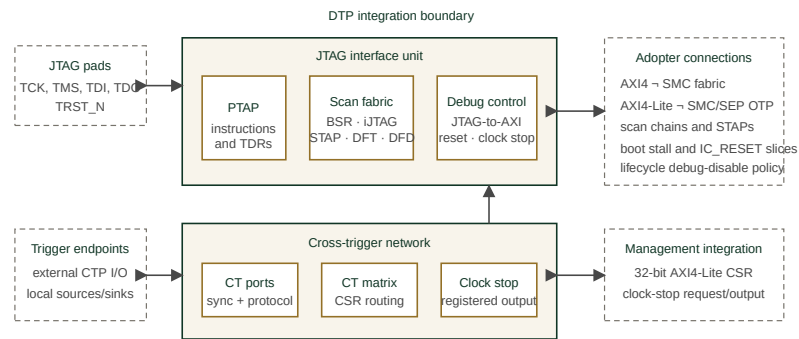
For SoC and chiplet teams that need a reusable test-access hub, controlled memory-mapped debug, and coordinated triggers across die boundaries. DTP is integrated in the OCAH SMU reference subsystem and can also serve as a subsystem-level integration boundary when its lifecycle, fabric, reset, clock, scan, and pad connections are supplied by the adopter.

OVERVIEW

OCAH DTP brings scan access, debug transactions, cross-trigger routing, and debug clock control together at one explicit integration boundary. A primary JTAG TAP selects boundary-scan, iJTAG, secondary-TAP, debug-control, capability, and JTAG-to-AXI data registers. Optional bridges expose the SMC system fabric and the SMC and SEP OTP interfaces to an authorized debugger. Active-high lifecycle disable fields are synchronized into the scan domain and gate the corresponding paths; the adopter remains responsible for generating the policy inputs and for securing downstream address maps.

The cross-trigger network connects external Cross Trigger Ports (CTPs) and chiplet-local trigger sources and sinks through a software-controlled matrix. It supports pulse and four-phase request/acknowledge signaling and combines debug and chiplet-local requests into a registered clock-stop output. Compile-time controls let adopters remove selected instructions, TAPs, bridge paths, and IC_RESET slices.

ARCHITECTURE



AT A GLANCE

Item	Checked-in DTP configuration
External / internal triggers	16 CTPs / 10 internal interfaces
Clock-stop request inputs	9
Cross-trigger CSR port	32-bit address, 32-bit data AXI4-Lite
SMC debug manager	56-bit address, 64-bit data AXI4
OTP debug managers	32-bit address, 32-bit data AXI4-Lite
Primary instruction register	6 bits

CAPABILITIES

JTAG and scan

- Primary TAP with IDCODE, BYPASS, SAMPLE/PRELOAD, and EXTEST
- Compile-time optional INTEST, CLAMP, HIGHZ, RUNBIST, EXTEST_PULSE, EXTEST_TRAIN, TMP, and CLAMP hold/release functions
- Boundary-scan host interface plus separate DFD, secure DFT, non-secure DFT, extended iJTAG, I/O STAP, SMC STAP, SEP STAP, and extra-STAP connections
- Capability registers report enabled functions and JTAG-to-AXI limits

Debug and control

- Single and series JTAG-to-AXI transactions, including incrementing and fixed-address series operations and target error reporting
- Independent lifecycle disable fields for three bridge paths and eight scan-side paths
- Typed IC_RESET slices for SMC, SEP, and adopter logic
- Debugger-controlled boot-stall and functional clock-stop requests

Cross trigger

- Bidirectional external CTP signaling and chiplet-local trigger interfaces
- Software-controlled source/destination routing with point-to-point and wire-OR behavior
- Pulse synchronization or four-phase request/acknowledge mode per internal trigger interface

INTERFACES AND CONFIGURATION

Boundary	Integration-visible controls
Clocks and resets	System clock/reset, JTAG TCK/TRST_N, JTAG power-on reset, scan reset, test enable
Debug security	<code>dbg_disable_i</code> lifecycle disable structure
Scan	PTAP pins; BSR/iJTAG scan controls; I/O, SMC, SEP, and extra STAPs
Fabric	One AXI4 manager, two AXI4-Lite managers, one AXI4-Lite subordinate
Cross trigger	External pad controls, internal source/sink interfaces, clock-stop requests and output
Build-time	Instruction/path enables, extra STAP count, IDCODE/version fields, IC_RESET types, trigger protocol modes, AXI type parameters

INTEGRATION DEPENDENCIES

- Generate lifecycle debug-disable policy outside DTP and connect the current active-high `dbg_disable_i` contract.
- Replace the zero-valued IDCODE manufacturer, part, revision, and OCH version defaults with release-specific values.
- Provide safe clock, reset, CDC, pad, and clock-gating integration. Establish project-specific TCK constraints; this beta sheet does not specify a characterized operating frequency.
- Configure IC_RESET slice types and enables, and connect or correctly bypass every enabled scan and secondary-TAP chain.
- Connect enabled AXI manager ports only to address maps permitted by the product security policy. Disabled and unused interfaces require documented tie-offs.
- Integrate the cross-trigger CSR subordinate into the management address map and connect the required local, external, and clock-stop endpoints.

VERIFICATION AND MATURITY

OCAH DTP is beta RTL. The repository’s open-source PyUVM environment exercises TAP operation, standard and optional instructions, JTAG-to-AXI data and error paths, backpressure/reset cases, iJTAG and STAP routing, cross-trigger modes, clock controls, and lifecycle-disable matrices. Scoreboards, reference models, coverage intent, and deliberately corrupted-checker tests guard against vacuous passes. A smaller SV-UVM scenario set is available for VCS.

No third-party standards certification, silicon characterization, or portable frequency, area, and power figures are claimed. Those results depend on the selected configuration, libraries, constraints, physical implementation, and target process.

DELIVERABLES AND FURTHER INFORMATION

- SystemVerilog RTL and Bender dependency manifests
- SystemRDL-derived cross-trigger register collateral
- Integration, architecture, interface, and verification-plan documentation
- Open-source cocoth/PyUVM tests, reusable JTAG/AXI agents, scoreboards, reference models, functional coverage, and fault-injection checks
- Synthesis and timing-constraint starting points
- Apache License 2.0 repository distribution

Technical detail: [hw/sys/dtp/doc/](#)
RTL: [hw/sys/dtp/rtl/](#)
Verification: [hw/sys/dtp/dv/](#)
Integrator Guide: [doc/integrator/](#)